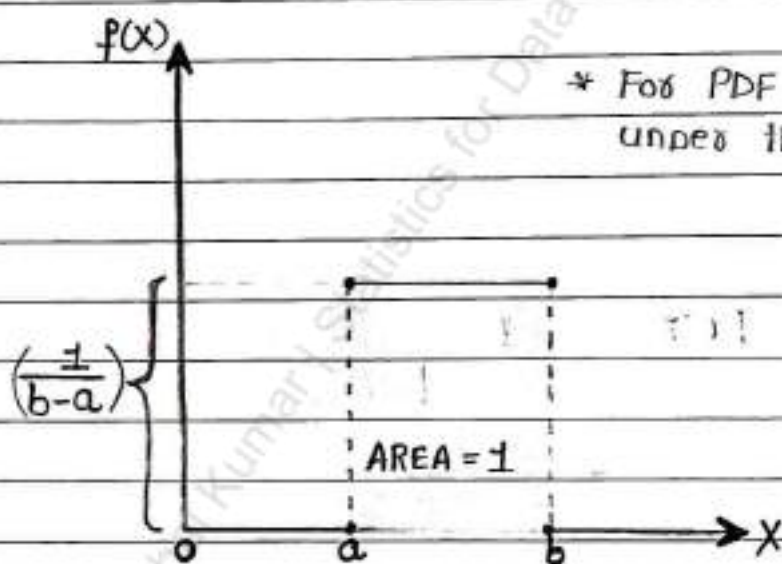


CONTINUOUS DISTRIBUTION.

I. UNIFORM DISTRIBUTION

- Uniform distribution is a continuous distribution that is characterized by constant probability density over a specified interval.
- Values in a uniform distribution are equally likely to occur.
- So if we plot a uniform distribution on chart:



•
$$\text{PDF} = f(x) = \frac{1}{b-a}, \text{ where } a \leq x \leq b$$

- Mean, $\mu = \frac{(a+b)}{2}$
- Variance = $\frac{(b-a)^2}{12}$

APPLICATIONS OF UNIFORM DISTRIBUTION

1. GUESSING A BIRTHDAY :

- If you pick a random person, each of the 365 days in a year is equally likely to be their birthday.
- Probability of any specific date = $(1/365)$

2. ROLLING A DIE :-

- A fair six-sided die has equal chance for each face.
- Probability for any number (1-6) = $1/6$.

3. RAFFLE TICKETS :-

- In a stadium raffle with 10,000 seats, each seat has the same chance of being chosen.
- Probability for any number/seat = $1/10,000$

4. DECK OF CARDS :-

- Drawing a random card, each suit (spades, hearts, clubs, diamonds) is equally likely.
- Probability for any suit = $(1/4)$

5. SPINNING A SPINNER

- A spinner divided into equal sections (red, green, blue) gives equal chance for each color.
- Probability for any particular color = $(1/3)$

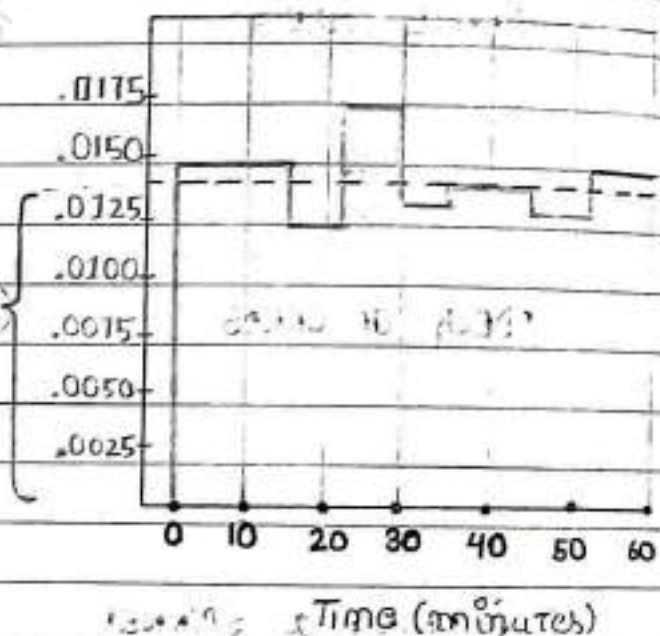
REAL-WORLD SCENARIO :-

⇒ Imagine you are a manager at a coffee shop, and you want to predict the arrival time of customers during a particular hour of operation. You believe that customers arrive at a relatively constant rate throughout the hour. To model this scenario, you can use a uniform distribution.

Solⁿ : $a=0$, $b=60$ as shop operates from 9:00 AM to 10:00 AM (60 min)

$$P(x) = \left(\frac{1}{60}\right) \text{ for all } x \text{ within this interval.}$$

→ Once we've drawn the rectangle representing the uniform distribution, the height of the rectangle which is $1/60$, reflects the equal likelihood of a customer arriving at any minute between 9:00 am - 10:00 AM.



→ The area under the curve represents the total probability, which is 1 — indicating that a customer will definitely arrive within that hour.

→ The grey rectangles in the graph represent's ⇒ HISTOGRAM
Each rectangle's height indicates the FREQUENCY/PROBABILITY of customer's arrivals within specific time intervals.

→ In this case, because it's a uniform distribution, the rectangles should ideally have the same height, reflecting that each time interval has an equal chance of customer arrival but the varying heights suggest that customer arrivals aren't evenly distributed throughout the hour, which deviates from a perfect uniform distribution.

* It's a real-world example, so some variations are expected.

In the graph, the taller grey bars around / within 20-30 minutes interval suggest a higher frequency of customer arrivals compared to other intervals, deviating from the uniform distribution where each minute would have an equal chance of arrival.

U.

II NORMAL DISTRIBUTION :-

→ The normal distribution → Gaussian Distribution, is a fundamental concept in probability theory and statistics.

It is a continuous probability distribution that is symmetric around its mean, indicating that data near the mean are more frequent in occurrence than data far from the mean.

* KEY CHARACTERISTICS :-

1. SYMMETRY :-

- The normal distribution is symmetric around its mean.
- This means the left side of the distribution mirrors the right side.

2. MEAN, MEDIAN and MODE :

In a normal distribution the mean, median and mode are all equal and located at the centre of the distribution.

3. BELL SHAPED CURVE :

The curve is BELL-SHAPED, indicating that most of the observations cluster around the central peak and the probabilities for values farther away from the mean taper off equally in both the directions.

4. STANDARD DEVIATION :

The spread of the distribution is determined by the std. About 68% of the data falls within 1 standard deviation of the mean, 95% within 2 standard deviations and 99.7% within 3 standard deviations.

$$* \text{PDF} = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, -\infty < x < \infty$$

Equal Halves

→ MEAN =

• VARIANCE = σ^2

• STANDARD DEVIATION = σ

